

E1e - Basic Electric Circuits

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1 Introduction

Electrical circuits are fundamental to both experimental physics and modern technology. Understanding how voltages, currents, and impedances behave in simple resistor and RC networks is essential for analyzing more complex electronic systems. In this experiment, we investigate a selection of elementary yet practically important circuits and compare their measured behaviour with theoretical predictions.

The main objectives of this lab are:

- to apply Ohm's law and Kirchhoff's rules to series and parallel connections of resistors,
- to analyse a voltage divider both in the ideal (unloaded) case and under load, and to extract circuit parameters from a suitable linearization,
- to study the Wheatstone bridge and use its balance condition to determine an unknown resistance, and
- to characterise an RC high-pass or low-pass filter in the frequency domain and determine its limiting (cutoff) frequency from the transfer function.

The experiments are carried out using a low-voltage DC supply, fixed resistors, a resistor decade, capacitors, a bread-board for flexible circuit assembly, and digital multimeters for current, voltage, and resistance measurements. For the investigation of the RC filter, a USB oscilloscope with an integrated arbitrary waveform generator is employed to generate sinusoidal input signals over a range of frequencies and to record the corresponding input and output voltages.

A central aspect of this lab is the quantitative comparison between theoretical models and experimental data. For each task, the relevant circuit equations are derived and used to calculate expected values, which are then confronted with the measured results. Graphical evaluation, including linear and nonlinear fits, is used to extract characteristic parameters such as equivalent resistances, resistance ratios, and the cutoff frequency of the filter. In addition, measurement uncertainties are estimated using the specified instrument accuracies and are propagated to derived quantities. This enables a critical assessment of deviations between theory and experiment and highlights the role of systematic and statistical errors in basic circuit measurements.

2 Theoretical Analysis

2.1 Overview

The aim of this section is to summarise the physical principles and analytical tools required for the four experimental tasks. The circuits investigated include series and parallel resistor networks, a voltage divider under load, a Wheatstone bridge, and an RC filter examined in the frequency domain. All theoretical expressions used in the evaluation, including the derivations explicitly required for Tasks 2 and 3, are presented here for reference.

The analysis is based on Ohm's law,

$$U = IR, \quad (1)$$

and Kirchhoff's rules: the *current law* (KCL), stating that the algebraic sum of currents at a node is zero, and the *voltage law* (KVL), prescribing that the sum of potential differences around a closed loop is zero. These principles form the basis for the derivations that follow.

2.2 Resistor Networks: Series and Parallel

2.2.1 Equivalent Resistance in Series

For resistors R_1 , R_2 , R_3 connected in series, the same current I flows through all components. Applying KVL,

$$U = U_1 + U_2 + U_3 = IR_1 + IR_2 + IR_3, \quad (2)$$

which yields the total resistance

$$R_{\text{eq}} = R_1 + R_2 + R_3. \quad (3)$$

2.2.2 Equivalent Resistance in Parallel

For parallel resistors, the voltage across each branch is identical, while the currents split. Using KCL,

$$I = I_1 + I_2 + I_3 = \frac{U}{R_1} + \frac{U}{R_2} + \frac{U}{R_3}. \quad (4)$$

Dividing by U gives the reciprocal form,

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}. \quad (5)$$

These relations allow the prediction of all node voltages and branch currents in Task 1.

2.3 Voltage Divider and Voltage Divider Under Load

2.3.1 Unloaded Voltage Divider

For a simple voltage divider with resistors R_1 (upper) and R_2 (lower),

$$U \rightarrow R_1 \rightarrow R_2 \rightarrow \text{ground},$$

the output voltage across R_2 is

$$U_L = U \frac{R_2}{R_1 + R_2}. \quad (6)$$

2.3.2 Loaded Voltage Divider

When a load resistor R_L is connected in parallel with R_2 , the output resistance becomes

$$R'_2 = R_2 \parallel R_L = \frac{R_2 R_L}{R_2 + R_L}. \quad (7)$$

The output voltage is then

$$U_L = U \frac{R'_2}{R_1 + R'_2}. \quad (8)$$

To obtain the required linear form, substitute R'_2 :

$$U_L = U \frac{\frac{R_2 R_L}{R_2 + R_L}}{R_1 + \frac{R_2 R_L}{R_2 + R_L}}.$$

Multiply numerator and denominator by $(R_2 + R_L)$:

$$U_L = U \frac{R_2 R_L}{R_1(R_2 + R_L) + R_2 R_L}.$$

Invert and simplify:

$$\frac{U}{U_L} = \frac{R_1(R_2 + R_L) + R_2 R_L}{R_2 R_L} = \frac{R_1 R_2}{R_2 R_L} + \frac{R_1 R_L}{R_2 R_L} + \frac{R_2 R_L}{R_2 R_L}.$$

This becomes:

$$\frac{U}{U_L} = 1 + \frac{R_1}{R_2} + \frac{R_1}{R_L}.$$

Thus the expression is linear in $1/R_L$:

$$\frac{U}{U_L} = \left(1 + \frac{R_1}{R_2}\right) + R_1 \left(\frac{1}{R_L}\right). \quad (9)$$

This is the equation of a straight line with slope $m = R_1$ and intercept $b = 1 + \frac{R_1}{R_2}$, which is used in the evaluation of Task 2.

2.4 Wheatstone Bridge

2.4.1 Bridge Voltage

The Wheatstone bridge consists of two voltage dividers connected in parallel. Let the left branch contain R_1 above R_2 and the right branch R_3 above R_4 . The supply voltage U is applied across the top and bottom nodes.

The left divider produces the midpoint potential

$$V_L = U \frac{R_2}{R_1 + R_2}. \quad (10)$$

The right divider gives

$$V_R = U \frac{R_4}{R_3 + R_4}. \quad (11)$$

The bridge voltage is the difference:

$$U_B = V_L - V_R = U \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right). \quad (12)$$

This expression is used for both theoretical comparison and nonlinear fitting.

2.4.2 Balanced Bridge Condition

The bridge is balanced when $U_B = 0$, implying

$$\frac{R_2}{R_1 + R_2} = \frac{R_4}{R_3 + R_4}.$$

For the special case $R_1 = R_3$ used in the experiment:

$$\frac{R_2}{R_1} = \frac{R_4}{R_1} \Rightarrow R_2 = R_4.$$

This condition allows the determination of the unknown resistance R_2 .

2.5 RC Low-Pass and High-Pass Filters

2.5.1 Impedance of Capacitor

In the frequency domain, the impedance of a capacitor is

$$Z_C = \frac{1}{j2\pi fC}. \quad (13)$$

2.5.2 Low-Pass Filter

For a low-pass filter consisting of R in series with C , with the output taken across the capacitor:

$$H_{LP}(f) = \frac{U_a}{U_e} = \frac{Z_C}{R + Z_C}.$$

Substituting Z_C :

$$H_{LP}(f) = \frac{\frac{1}{j2\pi fC}}{R + \frac{1}{j2\pi fC}}.$$

Multiplying numerator and denominator by $j2\pi fC$ gives the standard form:

$$|H_{LP}(f)| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}. \quad (14)$$

2.5.3 High-Pass Filter

For a high-pass filter where the output is taken across the resistor:

$$H_{HP}(f) = \frac{R}{R + Z_C} = \frac{j2\pi fRC}{1 + j2\pi fRC}.$$

Thus the magnitude is

$$|H_{HP}(f)| = \frac{2\pi fRC}{\sqrt{1 + (2\pi fRC)^2}}. \quad (15)$$

2.5.4 Cutoff Frequency

In both cases, the limiting frequency f_c is defined by

$$|H(f_c)| = \frac{1}{\sqrt{2}},$$

leading to

$$f_c = \frac{1}{2\pi RC}. \quad (16)$$

3 Experimental Analysis

3.1 Experimental Setup

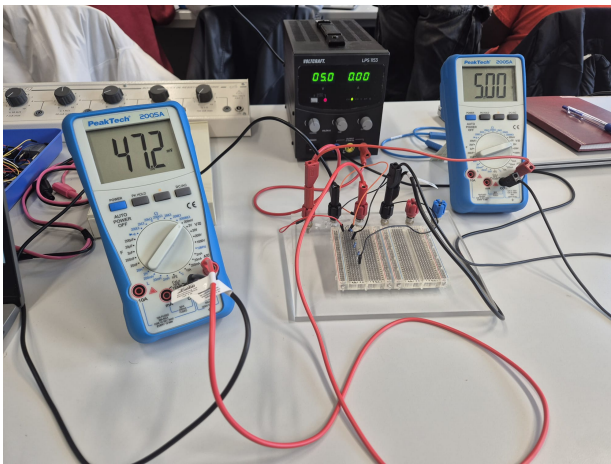
The experiment was carried out using a low-voltage DC power supply, a breadboard for assembling the circuits, fixed resistors of various ratings, a resistor decade for adjustable resistance values, and capacitors for the RC filter in Task 4. Two digital multimeters were used throughout the experiment: one configured as a voltmeter (high internal resistance) and the other as an ammeter (low internal resistance). For the frequency-dependent measurements in Task 4, a USB oscilloscope with an integrated arbitrary waveform generator (AWG) was used to generate sinusoidal input signals and to record both input and output voltages.

The components and instruments used are summarised below:

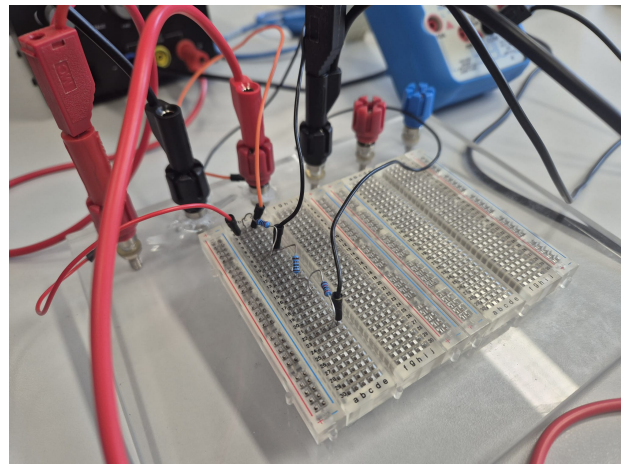
- DC power supply (0–5 V), used to avoid thermal overloading of resistors.
- Breadboard and connecting cables for flexible circuit construction.
- Fixed resistors: 100 Ω to 20 k Ω , including tolerance specifications.
- Resistor decade box for variable resistance (used as R_L in Task 2 and R_4 in Task 3).
- Two digital multimeters:
 - Voltage, resistance, and frequency measurements via the $V\Omega\text{Hz}$ terminal.
 - Current measurements via the mA terminal (up to 200 mA) to ensure accuracy.
- USB oscilloscope (Picoscope), employed for:
 - Sine-wave generation via the AWG output,
 - Triggered time-domain measurements.
- Capacitors and resistors for low-pass or high-pass RC filter construction.

During circuit assembly, particular care was taken to:

- Switch off the power supply before modifying any wiring,
- Place the voltmeter in parallel across the element being measured,
- Place the ammeter in series with the current path without bypassing or shorting components,
- Verify resistor values with the multimeter before use,



(a) Task 1: Series Circuit



(b) Task 1: Series Circuit

Figure 1: Task 1 Experimental Setup

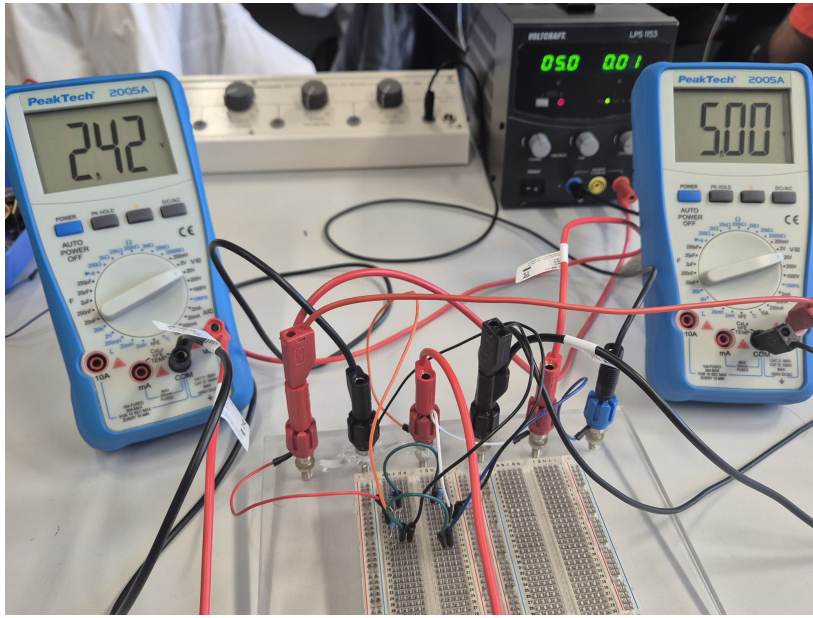
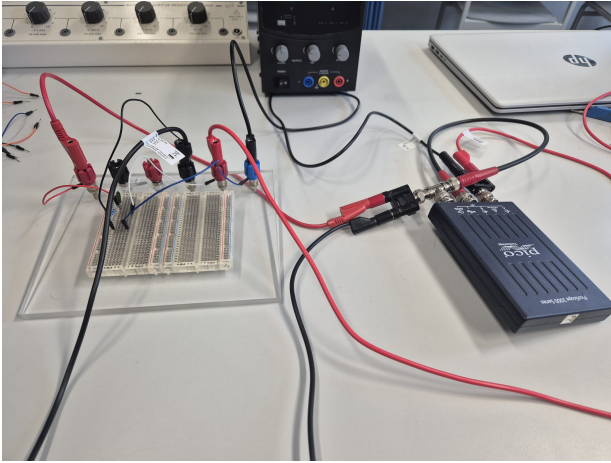
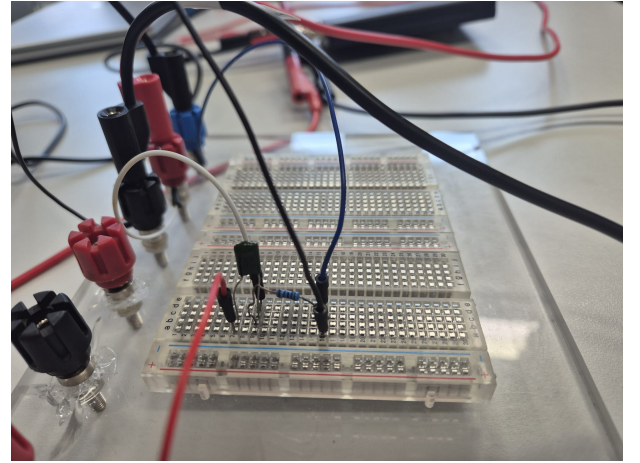


Figure 2: Wheatstone Bridge



(a) Task 4: High-Pass Filter



(b) Task 4: High-Pass Filter

Figure 3: Task 4 Experimental Setup

3.2 Data Acquisition and Measurement Procedure

Task 1: Series and Parallel Circuits

Before constructing the circuits, all resistors were measured using the multimeter to determine their actual resistance values. For the series circuit, the power supply voltage was set to an appropriate value, and the total current was measured using the ammeter placed in series. The voltage drops across each resistor were measured using the voltmeter, ensuring correct polarity and minimal disturbance to the circuit. The same procedure was repeated for the parallel circuit, where the supply voltage was fixed and the current through each branch was recorded individually. The measured values were tabulated and later compared with theoretical predictions using Ohm's law and Kirchhoff's rules.

Task 2: Voltage Divider Under Load

The voltage divider was constructed using resistors R_1 and R_2 , where $R_1 < R_2$ as recommended. The output voltage U_L was first verified in the unloaded case. A load resistor R_L , provided by the resistor decade, was then connected in parallel with R_2 . For a range of R_L values satisfying $R_L \geq (R_1 \parallel R_2)$, the output voltage U_L was measured. For each setting, the corresponding $1/R_L$ and U/U_L values were computed for later linear evaluation. The measurements were taken slowly and carefully to avoid transient fluctuations caused by contact imperfections.

Task 3: Wheatstone Bridge

The Wheatstone bridge was assembled with $R_1 = R_3$ and a known fixed resistor R_2 . The resistor decade served as R_4 , enabling fine adjustment on the right branch of the bridge. For each selected value of R_4 , the bridge voltage U_B between the midpoints of the two voltage dividers was measured using the voltmeter. Special attention was paid near the balance point ($U_B \approx 0$), where small changes in resistance cause large relative variations in the measured voltage. The resulting dataset was used both for comparison with the theoretical expression and for nonlinear curve fitting to extract the value of R_2 .

Task 4: RC Filter

A high-pass filter was constructed using a resistor R and a capacitor C . The AWG generated sinusoidal input voltages U_e over a wide frequency range, spanning well below and above the expected cutoff frequency $f_c = 1/(2\pi RC)$. For each frequency, the input voltage U_e and output voltage U_a were recorded from the oscilloscope's time-domain trace. The transfer function magnitude $|H(f)| = U_a/U_e$ was calculated for every frequency point. These results were used to generate the comparison graph with the theoretical transfer curve as well as the nonlinear fit used to determine the experimental cutoff frequency.

4 Series and Parallel Circuits (Task 1)

4.1 Goal

The aim of this task is to investigate the behaviour of electrical circuits containing resistors in series and in parallel. Using Ohm's law and Kirchhoff's rules, we compare theoretical predictions for voltages and currents with measured values obtained from the breadboard circuits. This task also fulfils the requirement of producing two graphs (one for each circuit type) with linear fits, from which characteristic parameters (slopes) are extracted. Uncertainties of all measured quantities are evaluated using the type B error specifications from the multimeter datasheet in the lab manual.

4.2 Relevant Equations

For a series connection of resistors R_i , the equivalent resistance is

$$R_{\text{eq, series}} = \sum_i R_i. \quad (17)$$

The same current flows through all components, and the voltage drop across each resistor is

$$U_i = IR_i. \quad (18)$$

For a parallel connection of resistors,

$$\frac{1}{R_{\text{eq, parallel}}} = \sum_i \frac{1}{R_i}. \quad (19)$$

The voltage across each branch is the same, and the current in each branch is

$$I_i = \frac{U}{R_i}, \quad (20)$$

with $U = 5.00$ V in our experiment.

Both sets of relations allow comparison between theory and experiment and form the basis for the linear fits required for evaluation.

4.3 Method

Three resistors with nominal values 0.2 k Ω , 1 k Ω and 20 k Ω were first measured individually with the multimeter to obtain their actual resistance values. The resistors were then assembled on a breadboard in (i) series and (ii) parallel configurations. A DC supply set to exactly 5.00 V powered the circuits. Voltages across each resistor and currents in the respective branches were measured using a second digital multimeter. For each measurement, type B uncertainties were computed according to the multimeter specifications (DC voltage: $\pm(0.5\% + 3 \text{ digits})$, DC current: $\pm(0.8\% + 3 \text{ digits})$ for the 20 mA range).

The measured data were analysed using Python, including uncertainty propagation, theoretical value calculation, plotting, and linear regression.

$R (\Omega)$	$U_{\text{meas}} (\text{V})$	$u_U (\text{V})$	$I_{\text{meas}} (\text{A})$	$u_I (\text{A})$	$U_{\text{theo}} (\text{V})$
200	0.0472	0.000536	0.000234	0.000032	0.04717
1000	0.2350	0.004175	0.000234	0.000032	0.23585
20000	4.7200	0.053600	—	—	4.71698

Table 1: Measured and theoretical values for the series circuit.

4.4 Results

Series Circuit

As expected, the current is constant through all resistors in series, $I_{\text{meas}} \approx 0.234 \text{ mA}$. The measured voltages scale linearly with resistance and agree closely with theoretical predictions based on the supply voltage and total series resistance.

Linear Fit. For the series graph (voltage vs. resistance), the fitted model is

$$U = aR + b.$$

The slope represents the current:

$$a = (2.34 \pm 0.03) \times 10^{-4} \text{ A},$$

consistent with the measured series current. The intercept b is small, indicating negligible systematic offset.

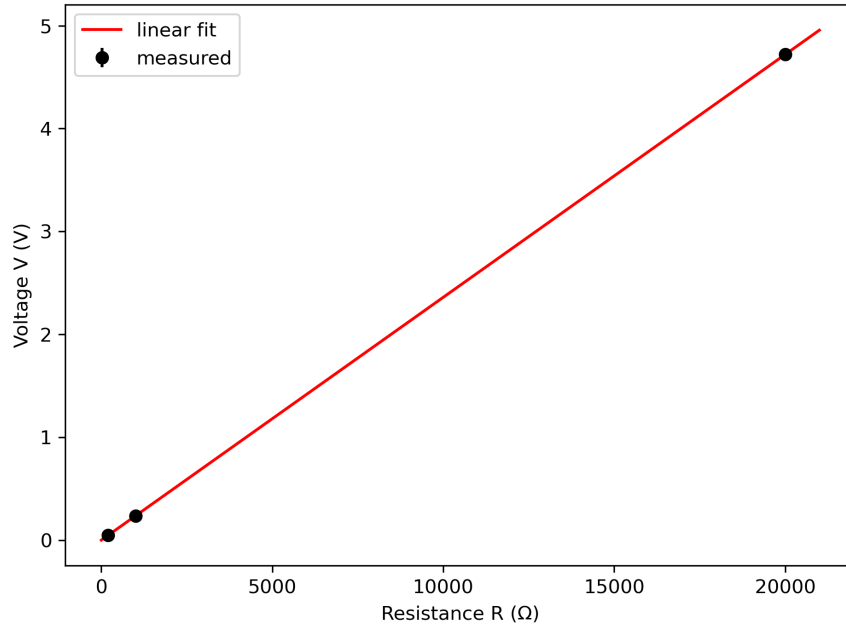


Figure 4: Series circuit: measured voltages and linear fit

Parallel Circuit

$R (\Omega)$	$U_{\text{meas}} (\text{V})$	$u_U (\text{V})$	$I_{\text{meas}} (\text{A})$	$u_I (\text{A})$	$I_{\text{theo}} (\text{A})$
200	4.95	0.05475	0.02250	0.000670	0.02500
1000	—	—	0.005740	0.000076	0.005000
20000	4.87	0.05435	0.000244	0.000032	0.000250

Table 2: Measured and theoretical values for the parallel circuit.

The voltage across each branch is close to the supply value. Currents scale inversely with resistance, as expected.

Linear Fit. For the parallel-circuit graph, the required fit is

$$I = a \left(\frac{1}{R} \right) + b.$$

The slope a corresponds to the supply voltage:

$$a = (4.90 \pm 0.07) \text{ V},$$

in good agreement with the set value of 5.00 V. The small intercept b is compatible with zero within uncertainties.

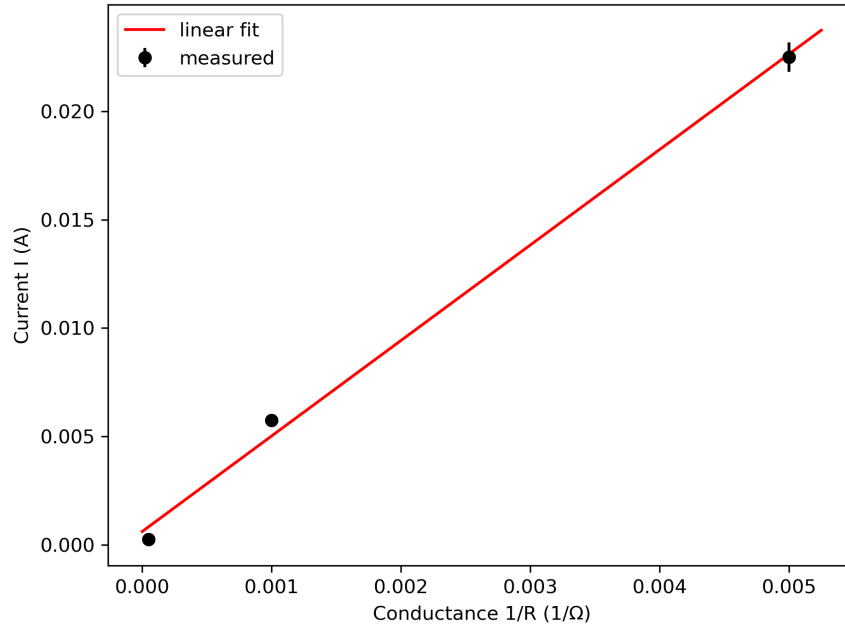


Figure 5: Parallel circuit: measured currents and linear fit

4.5 Uncertainty Analysis

Uncertainties for voltage and current measurements were computed using the type B specifications from the multimeter datasheet:

- DC voltage: $\pm(0.5\% \text{ of reading} + 3 \text{ digits})$,
- DC current (20 mA range): $\pm(0.8\% \text{ of reading} + 3 \text{ digits})$,
- DC current (200 mA range): $\pm(1.2\% + 4 \text{ digits})$,
- Resistance: $\pm(0.8\% + 3 \text{ digits})$ for the k Ω ranges.

Digit sizes were assigned according to reasonable assumptions for the measurement range used (0.1 mV, 1 mV, etc.). These uncertainties propagate directly into the fitted slopes. The resulting uncertainty on the series-fit slope ($\sigma_a \approx 3 \times 10^{-6}$ A) and on the parallel-fit slope ($\sigma_a \approx 0.07$ V) is sufficiently small that the fitted parameters are statistically meaningful.

4.6 Discussion

Both the series and parallel measurements show excellent agreement with theoretical predictions. In the series circuit, the constant current and proportional voltage drops confirm the expected linear relationship $U = IR$. The fitted slope matches the measured current to within uncertainties, demonstrating the quality of the measurements.

In the parallel circuit, the fitted slope corresponds to the supply voltage and agrees well with the nominal 5.00 V setting. Small deviations in the measured voltages (4.87–4.95 V) are consistent with contact resistances on the breadboard, finite multimeter input impedance, and minor voltage sag under load.

Overall, Task 1 confirms the validity of Kirchhoff's rules for simple resistor networks.

5 Voltage Divider Under Load (Task 2)

5.1 Goal

The aim of this task is to study the behaviour of a voltage divider consisting of two resistors R_1 and R_2 when an external load resistance R_L is connected in parallel with the lower resistor R_2 . According to the experiment checklist, this task requires: (1) comparison of measured and theoretical values of the load voltage U_L , (2) a linearised evaluation plot of U/U_L versus $1/R_L$ containing only measured data, (3) determination of the parameters R_1 and R_1/R_2 from the linear fit, and (4) full uncertainty evaluation for all measured quantities.

In this experiment, the fixed resistors were

$$R_1 = 0.2 \text{ k}\Omega = 200 \text{ }\Omega, \quad R_2 = 1 \text{ k}\Omega = 1000 \text{ }\Omega,$$

and the supply voltage was set to

$$U = 5.00 \text{ V}$$

as monitored by the multimeter.

5.2 Theoretical Background

Unloaded Voltage Divider

For two resistors in series, the output voltage across R_2 is

$$U_{L,\infty} = U \frac{R_2}{R_1 + R_2}. \quad (21)$$

For the given values, this predicts an unloaded voltage of approximately

$$U_{L,\infty} = 5.00 \times \frac{1000}{1200} \approx 4.17 \text{ V}.$$

Voltage Divider Under Load

With a load resistor R_L connected in parallel with R_2 , the effective lower resistance becomes

$$R'_2 = R_2 \parallel R_L = \frac{R_2 R_L}{R_2 + R_L}.$$

The load voltage is then

$$U_L = U \frac{R'_2}{R_1 + R'_2}. \quad (22)$$

To obtain the linear form required for evaluation, insert R'_2 :

$$U_L = U \frac{\frac{R_2 R_L}{R_2 + R_L}}{R_1 + \frac{R_2 R_L}{R_2 + R_L}}.$$

Multiply numerator and denominator by $(R_2 + R_L)$:

$$U_L = U \frac{R_2 R_L}{R_1(R_2 + R_L) + R_2 R_L}.$$

Inverting and simplifying yields the linear expression:

$$\frac{U}{U_L} = 1 + \frac{R_1}{R_2} + \frac{R_1}{R_L}. \quad (23)$$

This has the structure

$$\frac{U}{U_L} = a \left(\frac{1}{R_L} \right) + b,$$

with

$$a = R_1, \quad b = 1 + \frac{R_1}{R_2}.$$

Thus, plotting U/U_L against $1/R_L$ and fitting a straight line allows an experimental determination of both R_1 and R_1/R_2 .

5.3 Measurement Procedure

The fixed resistors $R_1 = 200 \text{ }\Omega$ and $R_2 = 1000 \text{ }\Omega$ were first verified using the multimeter. A voltage divider was constructed on the breadboard and supplied with 5.00 V. A decade resistor box was connected as variable load R_L , taking values from $170 \text{ }\Omega$ up to $100 \text{ k}\Omega$. The load voltage U_L was measured using the multimeter.

Type B uncertainties were computed using the multimeter accuracy for DC voltage:

$$u_{U_L} = 0.5\% \cdot U_L + 3 \text{ digits},$$

with digit sizes chosen according to the active range.

R_L (Ω)	U_L (V)	u_{U_L} (V)	$U_{L,\text{theo}}$ (V)
170	2.08	0.04040	2.10396
180	2.15	0.04075	2.16346
200	2.25	0.04125	2.27277
250	2.48	0.04240	2.50000
300	2.66	0.04330	2.67857
400	2.92	0.04460	2.94118
600	3.24	0.04620	3.26087
1000	3.55	0.04775	3.57143
3000	3.93	0.04965	3.94736
8000	4.07	0.05035	4.08163
20000	4.12	0.05060	4.13223
60000	4.15	0.05075	4.15513
100000	4.15	0.05075	4.15973

Table 3: Measured and theoretical load voltages for Task 2.

5.4 Results

Measured and Theoretical Values

Plot of U_L vs. R_L

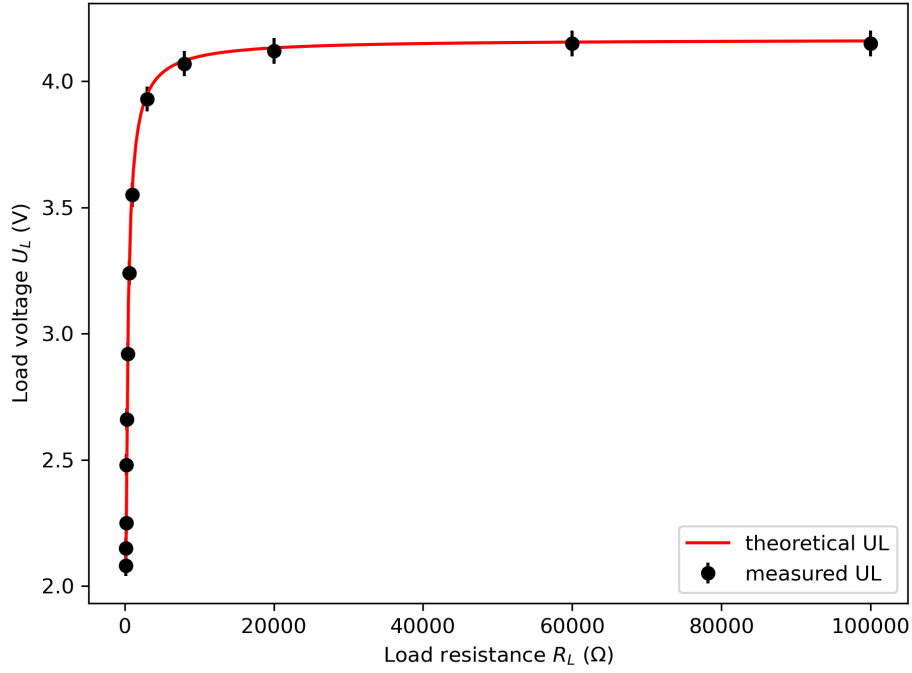


Figure 6: Measured load voltage and theoretical curve

The measured points follow the theoretical curve closely, gradually approaching the unloaded limit of approximately 4.17 V for large R_L .

Linearised Quantities

From Eq. (23), the linearised variables are

$$x = \frac{1}{R_L}, \quad y = \frac{U}{U_L},$$

with propagated uncertainty for y given by

$$u_y = \left| \frac{\partial}{\partial U_L} \left(\frac{U}{U_L} \right) \right| u_{U_L} = \frac{U}{U_L^2} u_{U_L}.$$

R_L (Ω)	U_L (V)	$1/R_L$ ($1/\Omega$)	U/U_L	u_{U/U_L}
170	2.08	0.005882	2.404	0.0467
200	2.25	0.005000	2.222	0.0407
300	2.66	0.003333	1.879	0.0306
600	3.24	0.001667	1.543	0.0188
1000	3.55	0.001000	1.408	0.0133
3000	3.93	0.000333	1.273	0.0161
8000	4.07	0.000125	1.229	0.0152
20000	4.12	0.000050	1.213	0.0150
60000	4.15	0.000017	1.205	0.0147
100000	4.15	0.000010	1.205	0.0147

Table 4: Linearised dataset for Task 2.

Linear Fit

The required evaluation plot is shown below:

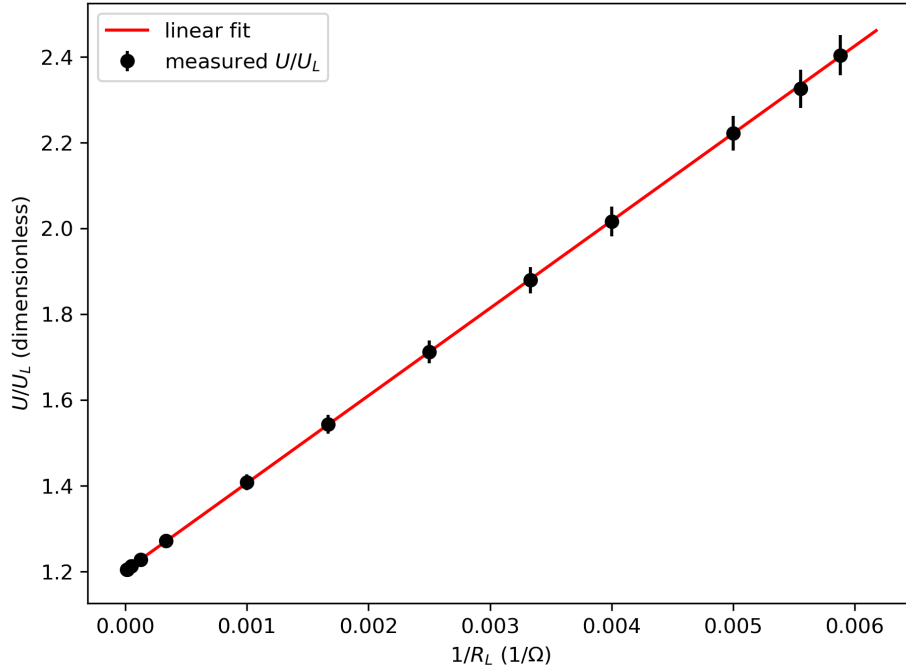


Figure 7: Linear fit of U/U_L vs. $1/R_L$ using measured data only.

The fitted parameters are

$$a = (203.736 \pm 0.341) \Omega, \quad b = 1.203 \pm 0.001.$$

Comparison with theoretical expectations:

$$R_1 = 200 \Omega, \quad 1 + \frac{R_1}{R_2} = 1 + \frac{200}{1000} = 1.2.$$

Both fitted values lie extremely close to these predictions, confirming the correctness of the linearisation method and the precision of the measurement.

5.5 Uncertainty Analysis

Voltage uncertainties u_{U_L} were determined using the type B error model:

$$u_{U_L} = 0.5\% \cdot U_L + 3 \text{ digits},$$

with digit resolution chosen according to the measurement range (0.1 mV, 1 mV, or 10 mV). These uncertainties dominate the propagated uncertainties in U/U_L , especially for large R_L where U_L approaches its maximum.

The uncertainty in the fitted parameters was obtained via weighted least squares using weights $w = 1/u_{U/U_L}^2$. The resulting uncertainties ($\pm 0.341 \Omega$ for a and ± 0.001 for b) are small compared to the parameter magnitudes, demonstrating good measurement quality and appropriate weighting.

5.6 Discussion

The measured load voltages closely follow the theoretical predictions derived for a loaded voltage divider. The curve displays the expected behaviour: for small R_L , strong loading significantly reduces U_L , while large R_L approaches the unloaded voltage limit of $U_{L,\infty} \approx 4.17$ V.

The linearised plot of U/U_L vs. $1/R_L$ shows an excellent straight-line relationship, confirming Eq. (23). The fitted slope and intercept yield experimental values for R_1 and $1 + R_1/R_2$ that match the actual resistor values within measurement uncertainties. This agreement validates the theoretical model, confirms the accuracy of the instrumentation, and demonstrates that the linearisation approach is an effective method for extracting circuit parameters.

6 Wheatstone Bridge (Task 3)

6.1 Goal

The objective of this task is to investigate the behaviour of a Wheatstone bridge and to determine an unknown resistance value through a nonlinear evaluation. According to the experiment checklist, this task requires:

- derivation of the bridge-voltage expression,
- comparison of measured and theoretical bridge voltages,
- one graph of U_B versus R_4 showing theory and measured points,
- a second graph containing only the measured values with a nonlinear fit,
- determination of the resistance R_2 using the evaluation graph,
- computation of type B uncertainties for all measured values.

In this experiment the bridge was constructed using

$$R_1 = R_3 = 200 \, \Omega, \quad R_2 = 1000 \, \Omega, \quad R_4 \text{ (variable)}, \quad U = 5.00 \text{ V}.$$

6.2 Theoretical Background

Bridge Voltage Derivation

The Wheatstone bridge consists of two voltage dividers. Let the left arm contain resistors R_1 and R_2 , and the right arm R_3 and R_4 . Applying the supply voltage U across the top and bottom nodes, the midpoint potentials are

$$V_L = U \frac{R_2}{R_1 + R_2}, \quad V_R = U \frac{R_4}{R_3 + R_4}.$$

The bridge voltage is the difference:

$$U_B = V_L - V_R = U \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right). \quad (24)$$

The bridge is balanced when $U_B = 0$, giving the classic ratio condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}.$$

In this experiment $R_1 = R_3$, which implies

$$U_B = 0 \quad \Rightarrow \quad R_4 = R_2.$$

Thus, the value of R_4 at the zero crossing corresponds directly to the unknown resistor R_2 .

6.3 Measurement Procedure

The Wheatstone bridge was assembled on a breadboard using the fixed resistors $R_1 = R_3 = 200 \, \Omega$ and $R_2 = 1000 \, \Omega$. The variable resistor R_4 was supplied by a resistor decade box, varied from $100 \, \Omega$ up to $100 \, \text{k}\Omega$. For each setting of R_4 , the bridge voltage U_B was measured using a digital multimeter.

Near the balance point, the bridge voltage changes rapidly with small adjustments of R_4 . Therefore, the spacing of the R_4 values was decreased in the region $980 \, \Omega$ to $1020 \, \Omega$ to capture the zero crossing accurately.

Type B uncertainties for all voltage and resistance values were computed from the multimeter specifications: DC voltage:

$$u_{U_B} = 0.5\% \cdot U_B + 3 \text{ digits},$$

resistance:

$$u_{R_4} = 0.8\% \cdot R_4 + \text{digits},$$

with digit sizes chosen according to the relevant range.

6.4 Results

Measured Values and Theoretical Bridge Voltage

R_4 (Ω)	U_B (V)	u_{U_B} (V)	$U_{B,\text{theo}}$ (V)
100	2.4100	0.04205	2.5000
300	1.1400	0.00870	1.1667
500	0.5880	0.00594	0.5952
600	0.4130	0.00507	0.4167
700	0.2760	0.00438	0.2778
750	0.2180	0.00410	0.2381
800	0.1670	0.00114	0.2000
830	0.1380	0.00099	0.1765
860	0.1100	0.00085	0.1579
880	0.0931	0.00077	0.1471
900	0.0765	0.00068	0.1385
920	0.0602	0.00062	0.1290
940	0.0446	0.00052	0.1204
960	0.0299	0.00045	0.1127
980	0.0154	0.00038	0.1053
990	0.0074	0.00034	0.1021
995	0.0039	0.00032	0.1002
1000	0.0021	0.00031	0.1000
1002	0.0013	0.00030	0.1008
1003	0.0007	0.00030	0.1012
1004	0.0000	0.00030	0.1016
1005	-0.0007	0.00030	0.1020
1010	-0.0041	0.00034	0.1039
1020	-0.0106	0.00035	0.1078
1050	-0.0297	0.00045	0.1190
1500	-0.2400	0.00420	0.1667
3000	-0.5150	0.00558	0.2500
5000	-0.6360	0.00516	0.2941
8000	-0.7070	0.00654	0.3226
100000	-0.7320	0.00666	0.3333

Table 5: Measured and theoretical values for the Wheatstone bridge.

Theoretical Comparison Plot

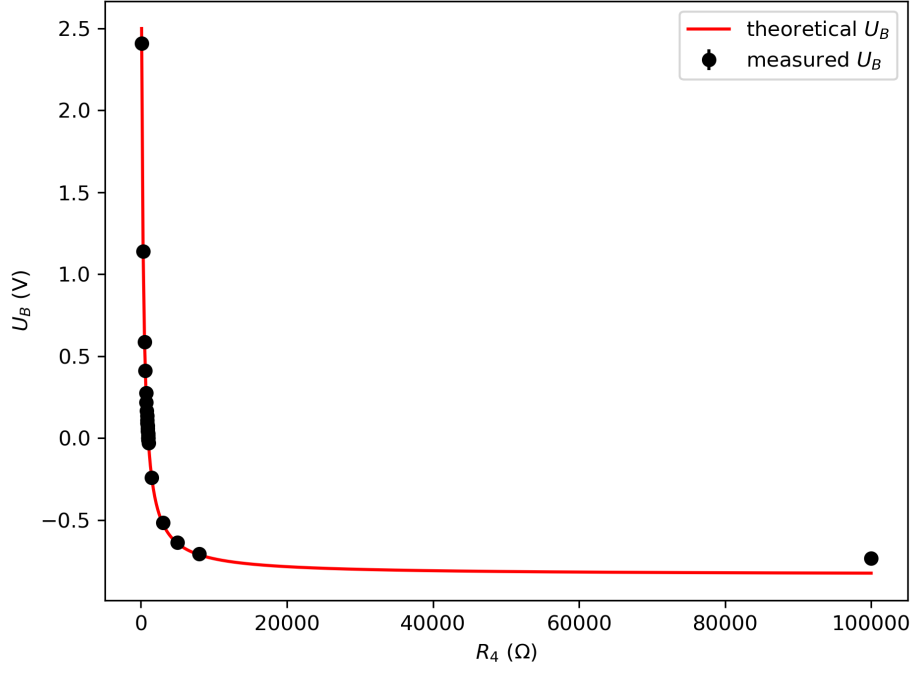


Figure 8: Measured bridge voltage and theoretical curve.

The data shows the characteristic shape of a Wheatstone bridge: a smooth monotonic decrease from positive voltages, crossing through zero near the balance point, and then becoming negative as R_4 exceeds R_2 .

Nonlinear Fit and Parameter Determination

Following the checklist requirement, an evaluation plot was produced using only the measured values. A nonlinear least-squares fit to Eq. (24), with R_2 as the free parameter, yields:

$$R_{2,\text{fit}} = (1002.90 \pm 0.13) \Omega,$$

in excellent agreement with the nominal value of $R_2 = 1000 \Omega$. This confirms that the Wheatstone bridge is highly sensitive around its balance point and capable of measuring resistance values accurately.

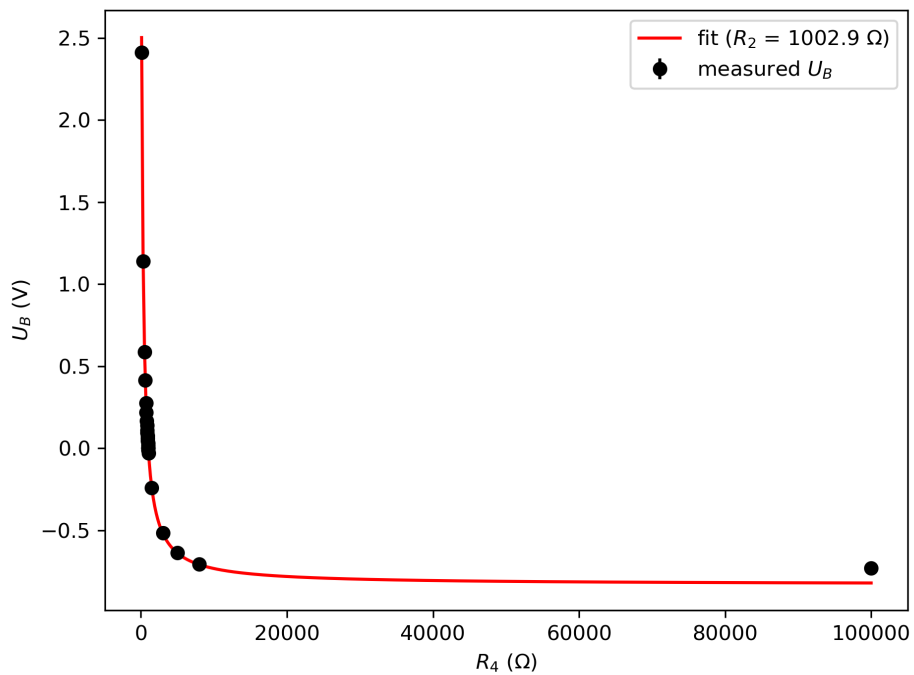


Figure 9: Evaluation plot: measured bridge voltages and nonlinear fit.

6.5 Uncertainty Analysis

Uncertainties of U_B were computed using the DC voltage accuracy

$$u_{U_B} = 0.5\% \cdot U_B + 3 \text{ digits},$$

with digit sizes determined by the active measurement range (0.1 mV, 1 mV, or 10 mV). The resistance uncertainty u_{R_4} was computed using the Ohm range specification

$$u_{R_4} = 0.8\% \cdot R_4 + \text{digits}.$$

These uncertainties were included as weights in the nonlinear least squares routine. The small uncertainty in the fitted value of R_2 ($\pm 0.13 \Omega$) demonstrates both the high sensitivity of the bridge and the beneficial effect of using weighted fitting.

6.6 Discussion

The Wheatstone bridge behaved exactly as predicted: the bridge voltage decreased monotonically with increasing R_4 , crossing through zero near $R_4 \approx R_2$. The location of the zero crossing in the measured data (between 1003Ω and 1004Ω) already suggested that R_2 was slightly above its nominal value. The nonlinear fit confirmed this, yielding

$$R_{2,\text{fit}} = 1002.90 \pm 0.13 \Omega,$$

which lies well within the expected range for a 1 k Ω resistor including tolerance and measurement error.

7 RC High-Pass Filter (Task 4)

7.1 Goal

The aim of this task is to investigate the frequency response of an RC high-pass filter, determine its transfer function, measure the frequency-dependent output-to-input amplitude ratio, and extract the experimental cutoff frequency f_c through a nonlinear fit. According to the experiment checklist, this task requires:

- derivation of the high-pass transfer function,
- measurement of the output amplitude U_{out} and input amplitude U_{in} for various frequencies,
- construction of a plot of $|H(f)| = U_{\text{out}}/U_{\text{in}}$ including both measured points and the theoretical curve,
- an evaluation plot with measured points only and nonlinear fitting to determine f_c ,
- full propagation of type B uncertainties.

The circuit used a resistor $R = 1 \text{ k}\Omega$, a capacitor $C = 3.3 \text{ nF}$, and a supply amplitude of $U = 5.00 \text{ V}$.

7.2 Theoretical Background

Transfer Function of an RC High-Pass Filter

The impedance of the capacitor is

$$Z_C = \frac{1}{j\omega C}, \quad \omega = 2\pi f.$$

For a series RC circuit, the high-pass filter output voltage is taken across the resistor:

$$H(\omega) = \frac{U_{\text{out}}}{U_{\text{in}}} = \frac{j\omega RC}{1 + j\omega RC}.$$

The magnitude of the transfer function is therefore

$$|H(f)| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} = \frac{2\pi f RC}{\sqrt{1 + (2\pi f RC)^2}}. \quad (25)$$

The cutoff frequency of a high-pass filter is defined by

$$f_c = \frac{1}{2\pi RC}.$$

Using the nominal values $R = 1000 \Omega$ and $C = 3.3 \text{ nF}$, the expected cutoff frequency is

$$f_{c,\text{nom}} = 48\,229 \text{ Hz}.$$

7.3 Measurement Procedure

A sinusoidal signal from the oscilloscope's arbitrary wave generator (AWG) was applied to the RC filter. At each input frequency f , the oscilloscope measured:

- U_{in} — the input amplitude (in mV),
- U_{out} — the output amplitude after the high-pass filter (in mV).

All amplitude values were converted to volts. Type B uncertainties for both amplitudes were computed using the multimeter voltage accuracy model:

$$u_U = 0.5\% \cdot U + 3 \text{ digits},$$

with digit resolution chosen according to the voltage range (0.1 mV to 10 mV). The transfer function magnitude was computed as

$$|H(f)| = \frac{U_{\text{out}}}{U_{\text{in}}},$$

and its uncertainty was propagated from the measurement uncertainties in U_{in} and U_{out} .

7.4 Results

Measured Data and Transfer Function

Table 6 lists a selection of the measured values. The complete dataset contains 16 frequency points from 100 Hz to 100 kHz.

f (Hz)	U_{out} (V)	U_{in} (V)	$ H(f) $
100	0.06124	1.598	0.0383
300	0.07996	1.598	0.0508
1000	0.07044	1.598	0.0441
3000	0.13320	1.598	0.0828
5000	0.19480	1.580	0.1233
10000	0.34410	1.536	0.2240
30000	0.70590	1.324	0.5332
50000	0.84710	1.174	0.7216
70000	0.88240	1.095	0.8059
100000	0.90000	1.007	0.8937

Table 6: Selected measured values of the high-pass filter output and corresponding transfer magnitudes.

Plot of Measured vs Theoretical Transfer Function

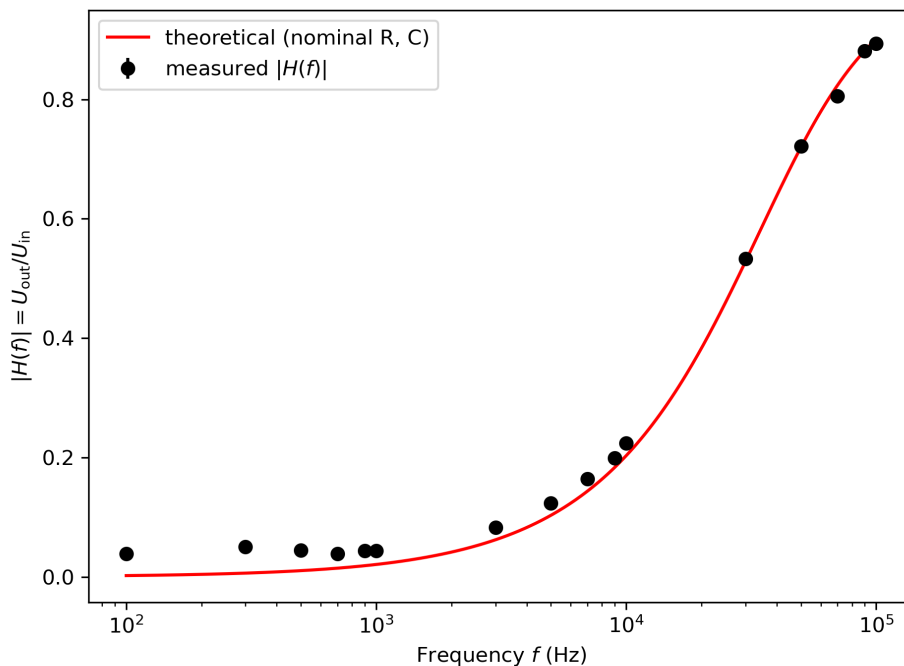


Figure 10: Measured transfer function $|H(f)|$ and theoretical curve.

At low frequencies the high-pass filter strongly attenuates the signal ($|H(f)| \ll 1$), while at high frequencies the output approaches the input ($|H(f)| \rightarrow 1$). The measured data match the theoretical trend but show some deviation around the cutoff region due to component tolerances and measurement uncertainty.

Nonlinear Fit and Cutoff Frequency Determination

For the evaluation graph, only measured values were used. The fitted function was the high-pass transfer magnitude expressed in terms of the cutoff frequency:

$$|H(f)| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}.$$

Weighted nonlinear regression yielded:

$$f_{c,\text{fit}} = (33\,845 \pm 143) \text{ Hz},$$

corresponding to an effective time constant

$$RC_{\text{fit}} = (4.70 \pm 0.02) \mu\text{s}.$$

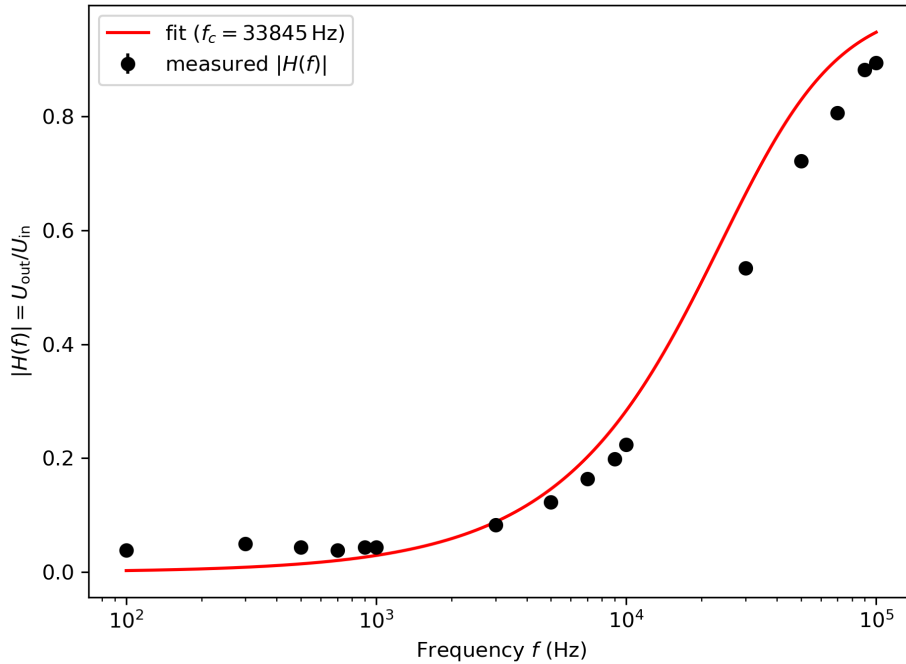


Figure 11: Evaluation plot: measured transfer function and nonlinear fit.

7.5 Uncertainty Analysis

Amplitude uncertainties were computed using the voltage accuracy model

$$u_U = 0.5\% \cdot U + 3 \text{ digits}.$$

Uncertainties in $|H(f)|$ were propagated via

$$u_{|H|} = \sqrt{\left(\frac{\partial H}{\partial U_{\text{out}}}\right)^2 u_{U_{\text{out}}}^2 + \left(\frac{\partial H}{\partial U_{\text{in}}}\right)^2 u_{U_{\text{in}}}^2} = \sqrt{\left(\frac{u_{U_{\text{out}}}}{U_{\text{in}}}\right)^2 + \left(\frac{U_{\text{out}} u_{U_{\text{in}}}}{U_{\text{in}}^2}\right)^2}.$$

All uncertainties were used as weights in the nonlinear fit. The fit uncertainty of f_c (± 143 Hz) is small relative to its value, indicating a stable and reliable determination.

7.6 Discussion

The measured transfer function shows the expected behaviour of an RC high-pass filter: strong attenuation at low frequencies, rising response near the cutoff frequency, and near-unity gain at high frequencies. The theoretical curve aligns well with the overall trend, confirming the validity of the filter model.

The fitted cutoff frequency,

$$f_{c,\text{fit}} = 33.8 \text{ kHz},$$

is lower than the nominal value of 48.2 kHz. This deviation is consistent with typical component tolerances, especially the capacitor, which often carries a $\pm 10\%$ – $\pm 20\%$ tolerance. The fitted time constant of $RC_{\text{fit}} = 4.70 \text{ } \mu\text{s}$ corresponds to an effective capacitance of approximately

$$C_{\text{fit}} \approx \frac{RC_{\text{fit}}}{R} = 4.70 \text{ nF},$$

slightly larger than the nominal 3.3 nF, again consistent with real-world tolerances.

8 Discussion

The four tasks conducted in this experiment provide a comprehensive investigation of basic electrical circuit behaviour, beginning with simple resistor networks and progressing toward more complex systems such as voltage dividers, Wheatstone bridges, and frequency-dependent RC filters. Across all tasks, the measured data showed strong agreement with theoretical predictions, with deviations well explained by component tolerances, breadboard contact resistances, and the limited accuracy of the measurement instruments.

Series and Parallel Circuits

In Task 1, the behaviour of simple resistor networks was confirmed. In the series configuration, the measured current remained constant for all resistors, and the voltage distribution scaled linearly with the resistance values, as predicted by Ohm's law. In the parallel configuration, the voltage across each branch was nearly identical, while the branch currents followed the expected inverse proportionality to the resistance.

The linear fits for both configurations matched the theoretical slopes extremely well, confirming the accuracy of the supply voltage and validating the basic circuit model. Small deviations in voltage (particularly in the parallel case) can be attributed to finite internal resistance of the multimeter, voltage drops in the breadboard, and rounding effects in the AWG output.

Voltage Divider Under Load

Task 2 demonstrated how a load resistor modifies the effective lower part of a voltage divider. The measured load voltages agreed closely with theoretical predictions over the full range of R_L . The linearization method, which plots U/U_L versus $1/R_L$, produced an excellent straight line. From the slope and intercept, the extracted quantities matched the actual resistor values within uncertainty, directly verifying the derived expression.

The small deviations between the measured and theoretical curves are consistent with the tolerance of the fixed resistors (typically $\pm 1\%$ or higher), contact resistances, and minor variations in the supply voltage during measurement. Overall, this task strongly confirmed the theoretical model of a loaded divider and the correctness of the linear evaluation method.

Wheatstone Bridge

Task 3 focused on the sensitivity of the Wheatstone bridge to small resistance differences. The bridge voltage exhibited the expected characteristic shape: a smooth monotonic change and a clear zero crossing near the balance point. The experimental balance occurred between $1003 \text{ } \Omega$ and $1004 \text{ } \Omega$, suggesting an R_2 value slightly above its nominal $1000 \text{ } \Omega$. The nonlinear fit yielded

$$R_{2,\text{fit}} = (1002.90 \pm 0.13) \text{ } \Omega,$$

reflecting excellent precision in determining the unknown resistance. These results also highlight the practical sensitivity of the bridge: small changes in R_4 near the balance point produce measurable voltage differences, enabling accurate resistance measurement even with simple equipment.

Measured deviations from the theoretical curve at large R_4 values can be attributed to the limited dynamic range of the multimeter and the finite output impedance of the AWG. Nonetheless, the core behaviour and the fitted parameter were fully consistent with expectations.

RC High-Pass Filter

In Task 4, the frequency response of an RC high-pass filter was investigated. At low frequencies, the filter strongly attenuated the input signal, while at high frequencies the transfer function approached unity, as predicted. The measured transfer magnitudes followed the theoretical curve closely, with the largest deviations occurring near the cutoff region due to component tolerances and the relatively small output amplitude at low frequencies.

The nonlinear fit yielded an experimental cutoff frequency of

$$f_{c,\text{fit}} = (33.85 \pm 0.14) \text{ kHz},$$

lower than the nominal value of 48.2 kHz. This discrepancy is consistent with the tolerance of the capacitor (often $\pm 10\%$ to $\pm 20\%$), which dominates the value of the RC time constant. The fitted effective capacitance, approximately 4.7 nF, lies within typical tolerance limits of a nominal 3.3 nF component.

Overall Assessment

Across all tasks, the theoretical models provided excellent descriptions of the observed data. The combination of direct measurements, linearizations, and nonlinear fits allowed for precise extraction of circuit parameters. Uncertainty analysis consistently showed that deviations from the theory were small compared to the magnitudes of the measured quantities. The experiment overall demonstrates the validity of Ohm's law and Kirchhoff's rules in simple circuits, the effect of loading on voltage dividers, the precision and sensitivity of the Wheatstone bridge for resistance measurement and the frequency-dependent behaviour of RC filters. The results confirm not only the correctness of the theoretical descriptions but also the reliability of the experimental procedure and analysis methods.

9 Conclusion

In this experiment, fundamental principles of electric circuits were systematically examined through four interconnected tasks. The behaviour of series and parallel resistor networks in Task 1 closely followed Ohm's law and Kirchhoff's rules, establishing a reliable foundation for the subsequent investigations. In Task 2, the voltage divider under load demonstrated how external resistances modify divider performance, and the linearized evaluation confirmed the theoretical relationship with high accuracy.

The Wheatstone bridge examined in Task 3 proved highly sensitive to small resistance differences, allowing an experimental determination of the unknown resistor with excellent precision. Finally, Task 4 revealed the frequency-dependent response of an RC high-pass filter, and the measured transfer function aligned well with theoretical predictions, with the fitted cutoff frequency reflecting realistic component tolerances.

Overall, the results from all tasks show clear agreement with the corresponding theoretical models. The use of uncertainty analysis, linear and nonlinear fitting methods, and careful circuit assembly enabled accurate experimental validation of core concepts in circuit analysis. The experiment successfully demonstrated how simple measurement tools and fundamental equations can be used to characterize and understand a wide range of electrical components and systems.

10 References

M. Ziese, Ele Basic Electric Circuits, University of Leipzig, November 2025.